Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Learner Registration Number

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Wednesday 10 January 2024

Afternoon (Time: 40 minutes)

Paper reference **31617H/1B**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
  – Section A: Structures and Functions of Cells and Tissues (Biology)
  – Section B: Periodicity and Properties of Elements (Chemistry)
  – Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

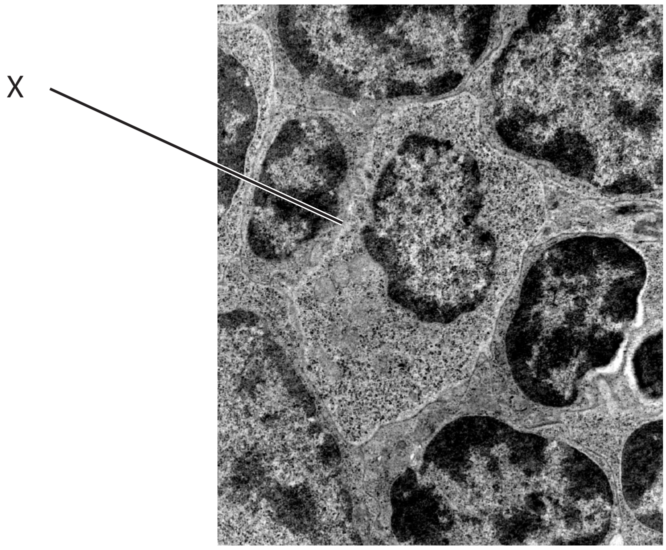
Turn over ▶

Pearson

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Figure 1 shows an electron micrograph of animal cells.



(Source: ©PAL)

Figure 1

The organelle labelled X is found in plant cells and animal cells.

(a) Identify the organelle labelled X in Figure 1.

(1)

☒ **A** cell wall

☒ **B** Golgi apparatus

☒ **C** nucleolus

☒ **D** plasma membrane

Plant cells and animal cells contain 80s ribosomes.

(b) State the function of 80s ribosomes.

(1)

....

(c) Identify the cell organelle that is found only in plant cells.

(1)

- ☒ **A** amyloplasts
- ☒ **B** cytoplasm
- ☒ **C** lysosomes
- ☒ **D** vesicles

Figure 1 has been produced using an electron microscope.

Some animal cells can be viewed using a light microscope.

(d) Explain **one** limitation of using a light microscope to view animal cells.

(2)

....

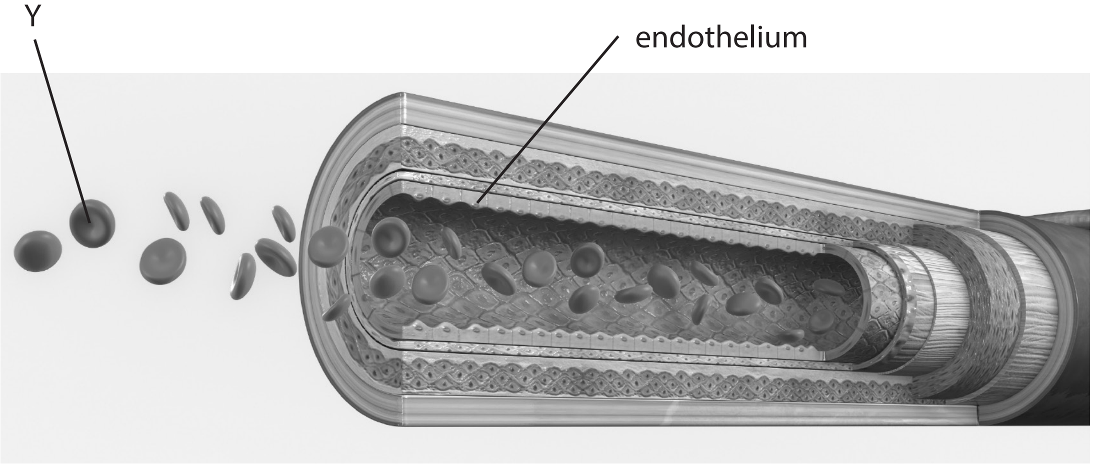
....

....

....

(Total for Question 1 = 5 marks)

2 (a) Figure 2 shows a diagram of part of a blood vessel.



(Source: ©PAL)

Figure 2

(i) Identify the cell labelled Y in Figure 2.

(1)

....

(ii) Explain the function of the endothelium.

(2)

....

....

....

....

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Paragraph 1 partially describes the development of atherosclerosis.

(iii) Complete Paragraph 1 by filling in the gaps to describe the development of atherosclerosis.

(3)

Atherosclerosis develops slowly with a build-up of cholesterol, fat, foam cells and other substances in the blood forming .... in the wall, under the endothelium.

The build-up causes the lumen to narrow.

The narrowing of the lumen increases the blood ....

This can cause more damage. This can lead to atherosclerosis.

Lifestyle choices, such as a poor diet, ... or lack of exercise can increase the risk of developing atherosclerosis.

Paragraph 1

(Total for Question 2 = 6 marks)

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3 Figure 3 shows a diagram of a myelinated neurone.

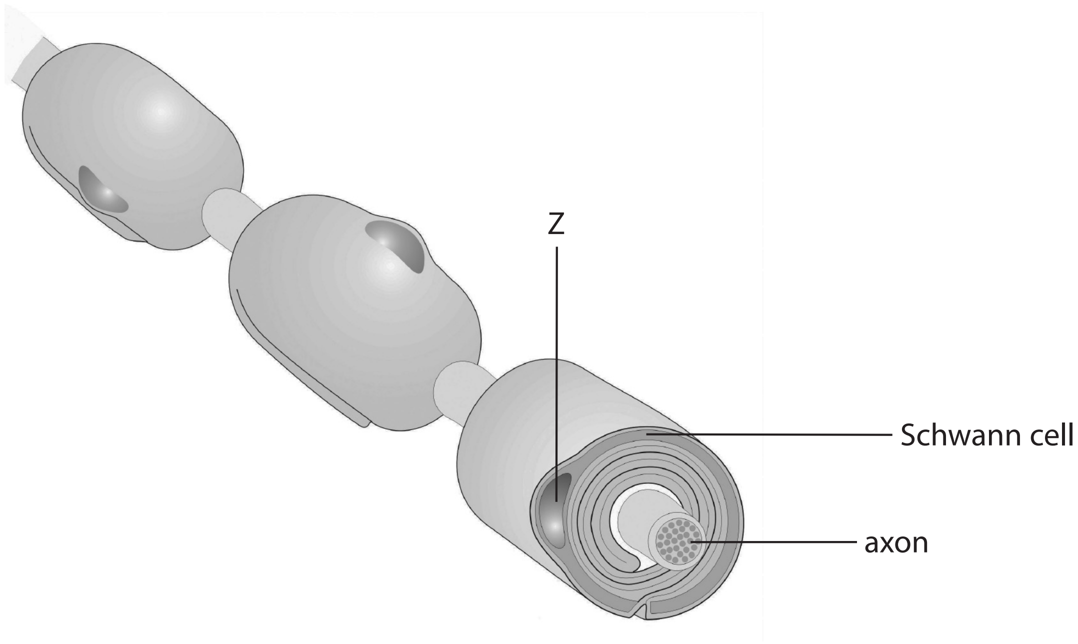

Figure 3

(Source: ©PAL)

(a) Identify the organelle labelled Z in Figure 3.

(1)

....

(b) Name the **two** ions involved in maintaining the resting potential in the myelinated neurone.

(2)

1 ....

2 ....

(c) The myelin sheath increases the speed of conduction of an impulse in a neurone.

The speed of conduction in a myelinated neurone is 40 m/s.

The speed of conduction in a non- myelinated neurone is 2 m/s.

(i) Calculate the percentage increase in the speed of conduction in a myelinated neurone compared with a non-myelinated neurone.

(3)

Show your working.

percentage increase = ... %

(ii) Name **one other** factor that affects the speed of conduction of an impulse.

(1)

....

(Total for Question 3 = 7 marks)

4 Figure 4 shows a diagram of a bacterium.

Bacterial pathogens can cause disease.

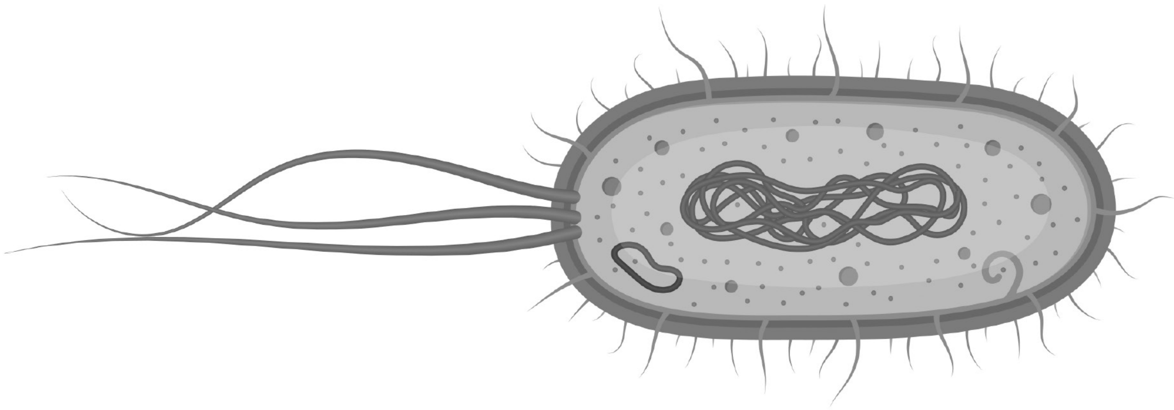

Figure 4

(Source: ©PAL)

The capsule and the plasmids are important to some bacterial pathogens.

The capsule protects the cell from drying out.

(a) Describe **two other** functions of the capsule in bacterial pathogens.

(2)

1 ....

....

2 ....

....

(b) Explain why plasmids are important for the survival of bacterial pathogens.

(4)

....

....

....

....

....

....

....

....

(Total for Question 4 = 6 marks)

5 Figure 5 shows a diagram of two root hair cells on the root of a plant.

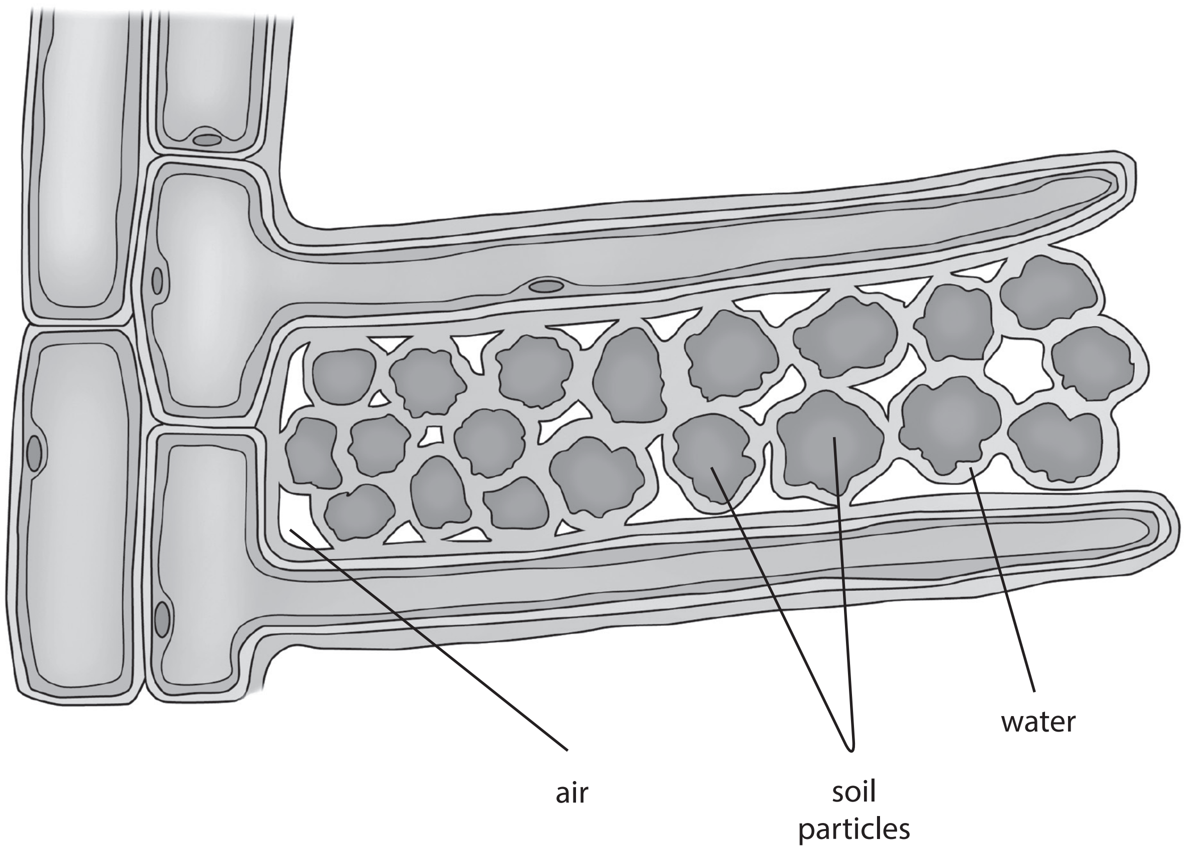

Figure 5

(Source: ©PAL)

A root hair cell has several functions, including absorption of water and ions.

Explain how the structural features of a root hair cell are specialised for the functions of the cell.

(6)

(Total for Question 5 = 6 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS

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